# Muntazir Ali Mughal

MSc Data Science student

muntaziralimughal6@gmail.com • +44 7943 424654 • London, United Kingdom • https://muntaziralim.github.io/

## PROFILE

MSc Data Science student at King's College London with a strong academic foundation in data analysis, machine learning, and applied software engineering. Experienced in delivering end-to-end data projects through coursework and personal projects, including data collection, feature engineering, modelling, and visualisation. Actively seeking graduate or junior analyst roles to apply data science skills in a professional environment while gaining real-world industry experience.

## EDUCATION

**MSc Data Science**, *King's College London* 09/2025 – Present

**BSc (Hons) Software Engineering**, *Liverpool John Moores University* 09/2022 – 09/2025

**BTEC Foundation Diploma**, *Pearson College* 09/2022 – 06/2023

## TECHNICAL EXPERTISE

| | | | |
|---|---|---|---|
| **Programming** | Python, Java, JavaScript | **Data Science** | Pandas, NumPy, Matplotlib, scikit-learn, Streamlit |
| **Machine Learning** | Supervised & Unsupervised Learning, Model Evaluation, XGBoost, SHAP | **GenAI / RAG** | LangChain, ChromaDB, Ollama, HuggingFace, RAG Pipelines |
| **MLOps** | MLflow, Evidently, DuckDB, SQL, Docker | **Databases** | SQL, Relational Databases, DuckDB, Data Handling |
| **Web** | HTML5, CSS3 | **Tools** | Git, VS Code, Power BI |

## SELECTED PROJECTS

**FinRisk Terminal — AI-Powered Financial Risk Intelligence Platform,** *Python, XGBoost, LangChain, RAG, DuckDB, ChromaDB, MLflow, Streamlit, Ollama* 03/2026

- Built an end-to-end financial risk platform combining XGBoost risk scoring (ROC-AUC: 0.68) with a RAG pipeline over 500+ real SEC 10-K filings using LangChain, ChromaDB and Ollama.
- Engineered a DuckDB SQL data warehouse ingesting 15,000+ rows of market data across 10 S&P 500 tickers with walk-forward time series validation to prevent data leakage.
- Implemented MLflow experiment tracking and Evidently AI drift monitoring to deliver a production-grade MLOps pipeline with full model observability.
- Deployed a Bloomberg-inspired Streamlit dashboard with an AI copilot answering natural language financial risk queries grounded in real SEC filings.
- Demonstrated 2026-relevant skills: RAG pipelines, LLM integration, MLOps, SQL data engineering, and end-to-end production deployment.

**Hospital Readmission Prediction,** *Python, XGBoost, SHAP, Streamlit* 02/2026 – 03/2026

- Built a machine learning model to predict 30-day hospital readmissions from clinical and demographic data.
- Tackled class imbalance using SMOTE/class weighting and evaluated models to select the best approach (XGBoost).
- Applied feature engineering and robust preprocessing to improve predictive signal and model stability.
- Used SHAP explainability to identify key drivers of readmission and explain individual predictions to stakeholders.
- Delivered an interactive Streamlit app to demo predictions and insights — demonstrating end-to-end delivery.

**Sign Language Translation Glove,** *Machine Learning & Sensors* 11/2024 – 07/2025

- Designed and built a wearable sensor-enabled glove capturing real-time hand movement data for gesture recognition.
- Integrated multi-sensor inputs to accurately detect finger positioning, motion, and orientation for reliable gesture tracking.
- Developed a machine learning pipeline to convert complex sign language gestures into real-time speech output.
- Trained and optimised motion-to-speech models to improve translation accuracy and responsiveness.
- Delivered an end-to-end working prototype showing accessible communication through assistive technology.

**Depot Management System – Inventory Optimisation,** *Full-Stack · Analytics* 12/2022 – 02/2023

- Architected and delivered a full-stack inventory management platform to automate depot planning and stock tracking.
- Implemented real-time inventory updates to improve visibility, reduce manual errors, and enhance stock accuracy.
- Automated reporting workflows to replace manual spreadsheets and reduce administrative workload.